



Adolescent Dietary Quality as a Risk Factor for Painful TMD & Headaches in Young Adults



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1 . Background

- **Painful TMD (P-TMD):**
 - The most common non-odontogenic-related chronic orofacial pain. ¹
 - Known risk factors: Gender, age, socio-economic status, and psychosocial factors
- P-TMD may overlap with primary **headaches**. ²
- **Dietary** behaviors and quality:
 - Among the most crucial lifestyle factors
 - Might impact chronic musculoskeletal pain conditions. ³

2 . Hypotheses

- Adolescent dietary habits predict the likelihood of developing P-TMD and headaches in young adulthood
- Adolescents’ intake of specific dietary domains influence the incidence of P-TMD and headaches in young adulthood

3 . Methods

- **Danish National Birth Cohort (DNBC)** ⁴
- **Dietary intake** – 36,064 individuals
 - 14-year-old participants
 - Healthy Eating Index (HEI) including eight dietary domains
- **P-TMD & Headache** – 33,412 individuals invited
 - Participants aged 18 years and above
 - Self-administered survey
- **Analyses** – 11,982 individuals responded
 - Logistic regression; dietary quality and P-TMD
 - Ordinal logistic regression; dietary quality and headaches
 - Adjusting for:
 - Sex, Socio-Occupational Status, Sleep quality, and Mental health
- ✓ **Model A:** Crude analysis
- ✓ **Model B:** Analysis adjusted for confounding factors with implementing inverse probability weighting (IPW)
- ✓ **Model C:** Analysis for individuals whose pain onset occurred after the measurement of dietary exposure (HEI)

4 . Results

Table 1. Distribution of P-TMD and Headache

		Healthy Eating Index (Higher quartile = Higher diet quality)				
		Quartile - 1	Quartile - 2	Quartile - 3	Quartile - 4	Total
P-TMD Number (%)	Negative	2,219 (74.21%)	2,152 (74.46%)	2,172 (73.88%)	2,140 (71.69%)	8,683 (73.55%)
	Positive	771 (25.79%)	738 (25.54%)	768 (26.12%)	845 (28.31%)	33,122 (26.45%)
	Total	2,990 (100%)	2,890 (100%)	2,940 (100%)	2,985 (100%)	11,805 (100%)
Headaches Number (%)	Less than 1 day/month	1,044 (36.36%)	979 (35.05%)	1,023 (36.42%)	1,033 (35.87%)	4,079 (35.93%)
	1 to 14 days/month	1,594 (55.52%)	1,558 (55.78%)	1,534 (54.61%)	1,586 (55.07%)	6,272 (55.25%)
	15 or more days/month	233 (8.12%)	256 (9.17%)	252 (8.97%)	261 (9.06%)	1,002 (8.83%)
	Total	2,871 (100%)	2,793 (100%)	2,809 (100%)	2,880 (100%)	11,353 (100%)

Table 2. Association of dietary quality at age 14 and P-TMD in young adulthood

	Odds Ratio (Model A)	P-Value	Odds Ratio (Model B)	P-Value	Odds Ratio (Model C)	P-Value
Healthy Eating Index (Overall)						
Quartile 2	0.98	0.82	0.95 (0.82-1.11)	0.57	0.94 (0.80-1.12)	0.53
Quartile 3	1.01	0.76	1.05 (0.90-1.23)	0.46	1.00 (0.85-1.18)	0.95
Quartile 4	1.13	0.02	1.12 (0.96-1.30)	0.11	1.06 (0.90-1.25)	0.41
Healthy Eating Index (domains)						
Fruit & Vegetables	1.01	0.009	1.01 (0.99-1.03)	0.07	1.00 (0.98-1.02)	0.41
Fiber	0.98	0.22	1.02 (0.99-1.04)	0.11	1.01 (0.98-1.04)	0.22
Fish	0.99	0.87	1.01 (0.99-1.03)	0.10	1.00 (0.98-1.02)	0.49
Total Meat	1.02	0.12	0.99 (0.96-1.03)	0.90	0.99 (0.96-1.03)	0.98
Saturated fatty acids	1.03	0.50	0.99 (0.87-1.13)	0.96	1.01 (0.88-1.16)	0.83
Sodium	1.02	0.009	0.99 (0.97-1.02)	0.95	1.00 (0.98-1.03)	0.57
Sugar-sweetened beverages	1.01	0.03	0.99 (0.97-1.01)	0.59	0.98 (0.96-1.01)	0.32
Added Sugar	1.01	0.18	1.01 (0.99-1.04)	0.11	1.01 (0.98-1.03)	0.27

Table 3. Association of dietary quality at age 14 and headache in young adulthood

	Odds Ratio (Model A)	P-Value	Odds Ratio (Model B)	P-Value	Odds Ratio (Model C)	P-Value
Healthy Eating Index (Overall)						
Quartile 2	1.07	0.16	0.99 (0.87-1.13)	0.94	0.98 (0.85-1.13)	0.81
Quartile 3	1.01	0.71	0.92 (0.80-1.05)	0.22	0.88 (0.76-1.02)	0.10
Quartile 4	1.04	0.43	0.91 (0.80-1.04)	0.19	0.90 (0.78-1.04)	0.18
Healthy Eating Index (domains)						
Fruit & Vegetables	1.00	0.14	0.98 (0.97-1.00)	0.17	0.98 (0.96-1.00)	0.16
Fiber	0.94	0.000	0.97 (0.94-0.99)	0.009	0.97 (0.95-0.99)	0.04
Fish	0.98	0.003	0.98 (0.97-1.00)	0.05	0.98 (0.97-1.00)	0.20
Total Meat	1.04	0.002	1.03 (0.99-1.07)	0.09	1.03 (0.99-1.07)	0.13
Saturated fatty acids	1.01	0.80	1.03 (0.80-1.20)	0.65	1.01 (0.87-1.18)	0.81
Sodium	1.05	0.000	1.03 (1.01-1.05)	0.004	1.03 (1.00-1.06)	0.01
Sugar-sweetened beverages	1.01	0.01	0.98 (0.97-1.00)	0.28	0.98 (0.96-0.99)	0.05
Added sugar	0.98	0.04	0.97 (0.95-0.99)	0.02	0.97 (0.95-0.99)	0.01

5 . Discussion

- Overall dietary quality in adolescence not significantly associated with P-TMD or headache in young adulthood after adjustment for confounders
- Specific dietary domains significantly associated with headache risk:
 - Fiber (protective), Added sugar (protective), Sodium
- Further exploring of pathophysiological links between lifestyle and P-TMD and Headache

6 . Take home

HEI's general influence on P-TMD and headaches is limited. Yet, specific dietary domains exhibit intriguing connections with headaches, requiring deeper exploration.

7 . References

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8 . Acknowledgements

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